RAJALAKSHMI ENGINEERING COLLEGE

Sripriya R 251801331-2025-AIDS S2

Started on	Tuesday, 25 August 2026, 1:01 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 1:10 PM
Time taken	9 mins 16 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
int n;
int i = 1;
int s = 1;
int count = 2;
scanf("%d",&n);
while(s<=n){
count++;
i++;
count++;
s+=i;
count++;
}
count++;
printf("%d",count);
return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Sripriya R 251801331-2025-AIDS S2

Started on	Wednesday, 26 August 2026, 11:20 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:23 AM
Time taken	2 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
int n;
int count = 0;
scanf("%d",&n);
count++;
if(n==1){
count++;
}
else{
    for(int i=1;i<=n;i++){
        count++;
        for(int j=1;j<=n;j++){
        count++;
        count++;
        count++;
        break;
    }
    count++;
}
count++;
}
printf("%d",count);
return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Sripriya R 251801331-2025-AIDS S2

Started on	Wednesday, 26 August 2026, 11:23 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:24 AM
Time taken	55 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
int n;
scanf("%d",&n);
int count = 1;
for(int i=1;i<=n;i++){
    count++;
    if(n%i==0){
        count++;

    }
    count++;

}
printf("%d",count);
return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Sripriya R 251801331-2025-AIDS S2

Started on	Tuesday, 25 August 2026, 1:08 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 1:10 PM
Time taken	1 min 39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int c = 0;
    c++;   // c = 0

    int i = n / 2;
    c++;   // i = n/2

    while (i < n)
    {
        int j = 1;
        c++;   // j = 1

        while (j < n)
        {
            int k = 1;
            c++;   // k = 1

            while (k < n)
            {
                c++;   // original c++

                k = k * 2;
                c++;   // k = k * 2
            }

            j = j * 2;
            c++;       // j = 2 * j
        }

        i++;
        c++;           // i++
    }
```

```
39
40      printf("%d", c);
41
42      return 0;
43  }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Sripriya R 251801331-2025-AIDS S2

Started on	Wednesday, 26 August 2026, 11:25 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:25 AM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int c = 0;
    c++;   // int c = 0

    int rev = 0, remainder;
    c++;   // rev = 0

    while (n != 0)
    {
        c++;   // while condition

        remainder = n % 10;
        c++;

        rev = rev * 10 + remainder;
        c++;

        n /= 10;
        c++;
    }

    c++;       // final while condition

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course